# PF LAB NO. 3

Name: Hamza Hasnain
Roll No: 48
Section: BS CS A
PF Lab No: 3

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## Written Work — Task 1 — Input statement

*Write one cin statement to read age, pay and section.*

### Code / Answer (write on A4 paper):

```
int age;
double pay;
char section;
cin >> age >> pay >> section;
```

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## Written Work — Task 2 — Expressions

*Evaluate the expressions using C++ integer arithmetic and operator precedence.*

### Code / Answer (write on A4 paper):

```
// Values in order:
28 / 4 - 2       // 5
6 + 12 * 2 - 8   // 22
4 + 8 * 2        // 20
6 + 17 % 3 - 2   // 5
2 + 22 * (9-7)   // 46
(8 + 7) * 2      // 30
(16 + 7) % 2 - 1 // 0
12 / (10 - 6)    // 3
(19-3)*(2+2)/4   // 16
```

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## Written Work — Task 3 — Implicit conversion

*Given int units; float mass; double weight; weight = mass * units; the product is promoted to float, then converted to double when assigned to weight.*

### Code / Answer (write on A4 paper):

```
int units = 4;
float mass = 2.5f;
double weight = mass * units;
cout << weight;
```

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## Written Work — Task 4 — Type conversion

*For int a, b = 2; float c = 4.2; a = b * c; the product is converted to int, so the fractional part is discarded.*

### Code / Answer (write on A4 paper):

```
int a, b = 2;
float c = 4.2f;
a = b * c;
cout << a;
```

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## Written Work — Task 5 — Avoid integer division

*Use an explicit cast so unitsEach can store a fractional result.*

### Code / Answer (write on A4 paper):

```
unitsEach = static_cast<double>(qty) / salesReps;
```

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## Written Work — Task 6 — Math tutor (random numbers)

*Display two random numbers, pause for the student, then show the correct sum.*

### Code / Answer (write on A4 paper):

```
#include <iostream>
#include <cstdlib>
#include <ctime>
using namespace std;
int main() {
srand(static_cast<unsigned>(time(nullptr)));
int a = rand() % 900 + 100;
int b = rand() % 900 + 100;
cout << a << "\n+ " << b << "\nPress Enter to see answer...";
cin.get();
cout << "\nAnswer = " << a + b << endl;
return 0;
}
```

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## Written Work — Home Task 1 — Fix the programs

*Corrected program to add two numbers (Program 1).*

### Code / Answer (write on A4 paper):

```
#include <iostream>
using namespace std;
int main() {
double number1, number2, sum;
cout << "Enter a number: ";
cin >> number1;
cout << "Enter another number: ";
cin >> number2;
sum = number1 + number2;
cout << "The sum of the two numbers is " << sum << endl;
return 0;
}
```

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## Written Work — Home Task 1 — Division program

*Corrected Program 2; casting prevents integer division.*

### Code / Answer (write on A4 paper):

```
#include <iostream>
using namespace std;
int main() {
int number1, number2;
double quotient;
cout << "Enter two numbers: ";
cin >> number1 >> number2;
if (number2 == 0) {
cout << "Cannot divide by zero.\n";
return 1;
}
quotient = static_cast<double>(number1) / number2;
cout << "Quotient = " << quotient << endl;
return 0;
}
```

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## Written Work — Home Task 2 — Average of five values

*Calculate the sum of 28, 32, 37, 24 and 33, then divide by 5.*

### Code / Answer (write on A4 paper):

```
#include <iostream>
using namespace std;
int main() {
int a = 28, b = 32, c = 37, d = 24, e = 33;
int sum = a + b + c + d + e;
double average = sum / 5.0;
cout << "Sum = " << sum << endl;
cout << "Average = " << average << endl;
return 0;
}
```

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## Laptop — Code Editor: Task 1 — Input statement

Code::Blocks / C++ source

```
int age;
double pay;
char section;
cin >> age >> pay >> section;
```

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## Laptop — Program Output: Task 1 — Input statement

```
Input: 19 25000 A
Values accepted.
```

*Illustrative terminal output — run the code on your laptop to capture the actual result.*

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## Laptop — Code Editor: Task 2 — Expressions

Code::Blocks / C++ source

```
// Values in order:
28 / 4 - 2       // 5
6 + 12 * 2 - 8   // 22
4 + 8 * 2        // 20
6 + 17 % 3 - 2   // 5
2 + 22 * (9-7)   // 46
(8 + 7) * 2      // 30
(16 + 7) % 2 - 1 // 0
12 / (10 - 6)    // 3
(19-3)*(2+2)/4   // 16
```

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## Laptop — Program Output: Task 2 — Expressions

```
Program started.
Enter sample values as prompted.
Program finished successfully.
```

*Illustrative terminal output — run the code on your laptop to capture the actual result.*

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## Laptop — Code Editor: Task 3 — Implicit conversion

Code::Blocks / C++ source

```
int units = 4;
float mass = 2.5f;
double weight = mass * units;
cout << weight;
```

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## Laptop — Program Output: Task 3 — Implicit conversion

```
2.5 x 4
10
```

*Illustrative terminal output — run the code on your laptop to capture the actual result.*

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## Laptop — Code Editor: Task 4 — Type conversion

Code::Blocks / C++ source

```
int a, b = 2;
float c = 4.2f;
a = b * c;
cout << a;
```

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## Laptop — Program Output: Task 4 — Type conversion

```
Output: 8
```

*Illustrative terminal output — run the code on your laptop to capture the actual result.*

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## Laptop — Code Editor: Task 5 — Avoid integer division

Code::Blocks / C++ source

```
unitsEach = static_cast<double>(qty) / salesReps;
```

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## Laptop — Program Output: Task 5 — Avoid integer division

```
Program started.
Enter sample values as prompted.
Program finished successfully.
```

*Illustrative terminal output — run the code on your laptop to capture the actual result.*

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## Laptop — Code Editor: Task 6 — Math tutor (random numbers)

Code::Blocks / C++ source

```
#include <iostream>
#include <cstdlib>
#include <ctime>
using namespace std;
int main() {
srand(static_cast<unsigned>(time(nullptr)));
int a = rand() % 900 + 100;
int b = rand() % 900 + 100;
cout << a << "\n+ " << b << "\nPress Enter to see answer...";
cin.get();
cout << "\nAnswer = " << a + b << endl;
return 0;
}
```

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## Laptop — Program Output: Task 6 — Math tutor (random numbers)

```
Math Tutor
 348
+ 527
Answer = 875
```

*Illustrative terminal output — run the code on your laptop to capture the actual result.*

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## Laptop — Code Editor: Home Task 1 — Fix the programs

Code::Blocks / C++ source

```
#include <iostream>
using namespace std;
int main() {
double number1, number2, sum;
cout << "Enter a number: ";
cin >> number1;
cout << "Enter another number: ";
cin >> number2;
sum = number1 + number2;
cout << "The sum of the two numbers is " << sum << endl;
return 0;
}
```

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## Laptop — Program Output: Home Task 1 — Fix the programs

```
Enter a number: 12
Enter another number: 8
The sum of the two numbers is 20
```

*Illustrative terminal output — run the code on your laptop to capture the actual result.*

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## Laptop — Code Editor: Home Task 1 — Division program

Code::Blocks / C++ source

```
#include <iostream>
using namespace std;
int main() {
int number1, number2;
double quotient;
cout << "Enter two numbers: ";
cin >> number1 >> number2;
if (number2 == 0) {
cout << "Cannot divide by zero.\n";
return 1;
}
quotient = static_cast<double>(number1) / number2;
cout << "Quotient = " << quotient << endl;
return 0;
}
```

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## Laptop — Program Output: Home Task 1 — Division program

```
Enter two numbers: 7 2
Quotient = 3.5
```

*Illustrative terminal output — run the code on your laptop to capture the actual result.*

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## Laptop — Code Editor: Home Task 2 — Average of five values

Code::Blocks / C++ source

```
#include <iostream>
using namespace std;
int main() {
int a = 28, b = 32, c = 37, d = 24, e = 33;
int sum = a + b + c + d + e;
double average = sum / 5.0;
cout << "Sum = " << sum << endl;
cout << "Average = " << average << endl;
return 0;
}
```

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## Laptop — Program Output: Home Task 2 — Average of five values

```
Sum = 154
Average = 30.8
```

*Illustrative terminal output — run the code on your laptop to capture the actual result.*